Modern experimental mechanics FHNL10

Learning outcomes

Knowledge and understanding

For a passing grade the student must

- be familiar with the different modern experimental techniques, and in particular full-field measurements, for studying the mechanics of solid, porous and granular materials
- understand the concepts of digital image correlation approaches and experimental set-ups
- understand the basics of neutron and x-ray techniques applied to experimental mechanics
- understand the basics of wave propagation and wave measurements in experimental mechanics

Competences and skills

For a passing grade the student must

- be able to formulate and carry out simple digital image/signal processing procedures
- be able to interpret and discuss results from different experimental techniques
- be able to formulate and solve simple inverse problems in experimental mechanics

Judgement and approach

For a passing grade the student must

- be able to propose appropriate experimental methods to answer a given research question
- be able to assess sources of error in experimental analyses
- be able to appreciate the limitations of different experimental methods

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Assessment:

- The course assessment involves 4 short practical assignments during the course plus a summary written assessment submitted at the end of the course.
- These assignments all contribute to the final course grade – **all assignments must be submitted** and an aggregate grade is given based on all the submitted assignments
- Final course grade is given as failed, 3, 4, 5.
- Procedures for extra assignments or revision of assignments, when failed, are arranged after contact with the course coordinator.

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Assessment: Experiment reports

- The course involves 4 “experiment” assignments: Tensile testing, Digital Image Correlation, X-ray tomography analysis and X-ray scattering analysis

- Each of the “experiment” assignments should be written up as an experiment report such that sufficient details are given for the reader to be able to repeat the experiment and analysis including understanding the basis of any interpretations or conclusions drawn from the results. This includes providing any (matlab®/python) code used in the analysis.

- **Key aspects to each assessment will be:**
  - Presentation of the report (structure, readability, citations, figure captions and referencing, etc.)
  - Description of the experiment, including key parameters
  - Description of the analysis approach linked to the underlying physics
  - Success in the proposed tasks (basic analysis up to extended analysis)
  - Presentation of results
  - Discussion of results and conclusions drawn
  - Analysis of method and recognition of limitations and identification of improvements / further possible work

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Assessment: Final Dossier

- The “course dossier” is a written summary of the course content submitted at the end of the course.
- There are 6 “subjects” in total and each one should be summarised in 2-4 pages
- The summaries can include figures/images, plus should include references to all utilised sources and should be written in a report form (i.e., not just bullet points)
- The aim is to demonstrate your understanding of the subject and to demonstrate any further investigation beyond the lecture content (this could include what has been learnt from the experiment/analysis assignments or from your own literature searches)
- A pass requires all the different “guiding questions” to be addressed in some detail demonstrating understanding of the method (underlying physics, measurement/analysis approach(es), practical aspects and possible applications).
- Higher grades are awarded to summaries developed beyond just answering the guiding questions and including content beyond the lecture content, including, e.g., examples of applications of the different methods found in the literature (with correct citation). Discussion of the limitations and advantages of the methods as well as sources of possible error/uncertainty are also key.

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Assessment: Final Dossier

Subject summaries

- Short (2-3 pages) summary of each topic
- Objective is to gain a greater depth of understanding of the subject through self-study and to have a set of **summary** notes
- These summaries will be used to assess the understanding of each topic
- Resources: course notes and literature from the website, web searches (google, Wikipedia), suggested references, etc.
- Guiding questions will be provided (but don't limit yourselves to these!)

- Subject summaries:
  - Experimental method in mechanics and full-field methods
  - Digital image correlation for Experimental Mechanics
  - Acoustics, Ultrasonics & Ultrasonic tomography for Experimental Mechanics
  - X-ray scattering for Experimental Mechanics
  - X-ray tomography for Experimental Mechanics
  - Neutrons for Experimental Mechanics

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Points distribution:

1 – <u>Final "dossier"</u>: 18 points

2 – <u>Experiment reporting</u>: 42 points

- **Lab 1 report -** 2 points
- **DIC project report -** 18 points
- **X-ray Tomography project report -** 10 points
- **WAXS project report -** 12 points

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Guidelines and working practice

- **All assessments should be submitted individually**
- Discussion and group work is allowed (and encouraged), but final submissions should be written separately and not contain the same text.
- **References** should be given to literature, websites, etc., used (you are encouraged to read around the subject)
- Matlab®/python code should be your own (do not be tempted to copy from the web / previous years)
- Matlab® can be installed through the university or can be used online with your university login
- You are also welcome to work with Python if you would rather, but the exercises are currently focused towards Matlab®
- The exercise classes / help sessions are timetabled to get help – please make use of these before coming to us out-of-hours
- Of course, ChatGPT etc. exist as a tools, but be aware that you should not trust any single source and the output must be verified and you must understand it plus give references and you own perspectives…
- …Likewise, LLMs can be used to accelerate your code development, but you need to understand what the code is doing and use the code to produce meaningful analysis of your data.